

Minimal Structural Sufficiency and the Pruning Paradox in Neural Networks

Abstract

Modern neural networks exhibit extreme parameter redundancy: up to 90% of weights can be removed without immediate loss of performance. This empirical fact is well documented through the Lottery Ticket Hypothesis, structured sparsity, and hardware-aware pruning techniques. However, performance does not degrade smoothly with pruning. Instead, it remains stable across wide removal regimes and then collapses abruptly.

This work argues that sparsity alone cannot explain this behavior. We introduce the concept of **minimal structural sufficiency** (F_{min}): the minimal, non-distributable configuration required for stable inference. Pruning succeeds not because models are sparse, but because it preserves F_{min} . Failure occurs when pruning violates this structural threshold, triggering abrupt throughput collapse. This framework provides a unifying explanation for pruning dynamics, failure modes, and efficiency limits in minimal-resource machine learning.

1. Introduction — The Pruning Paradox

Several empirical facts about modern neural networks are broadly uncontested:

- Overparameterized networks contain massive redundancy.
- Large fractions of parameters can be removed with little or no loss in accuracy.
- Structured sparsity enables significant inference speedups.
- Performance remains stable across wide pruning ranges.

Yet an unresolved problem persists:

Why does performance remain stable until it suddenly collapses?

Existing pruning literature describes *how* parameters can be removed efficiently, but does not explain *what must remain* for inference to remain viable. Accuracy curves do not degrade proportionally to parameter count. Instead, they exhibit plateau-like behavior followed by sharp transitions.

This paper argues that pruning behavior is governed not by sparsity levels, but by the preservation or violation of a minimal structural core.

2. Limits of Sparsity-Based Explanations

Sparsity is often treated as a proxy for efficiency. However, sparsity alone is insufficient as an explanatory principle.

- Two networks with identical sparsity ratios can exhibit radically different performance.
- Hardware-aware pruning improves execution speed but does not explain inference stability.
- Magnitude-based criteria correlate weakly with functional importance.

Sparsity measures *how much* is removed, but says nothing about *what cannot be removed*.

The key variable is not parameter count, density, or compression ratio, but **structural sufficiency**.

3. Minimal Structural Sufficiency (F_{min})

We define **minimal structural sufficiency**, denoted F_{min} , as:

■ *The minimal structural configuration required for stable inference on a given task.*

Properties of F_{min} :

- **Non-distributable**
It cannot be arbitrarily spread across parameters; certain structures are irreducible.
- **Non-recoverable once breached**
Once violated, retraining cannot reliably restore performance.
- **Architecture-relative but task-invariant**
Different architectures may realize different F_{min} , but the task imposes a fixed structural requirement.
- **Independent of parameter count**
Large and small models may share the same effective F_{min} .

Pruning succeeds because it removes non-contributing substructures while preserving F_{min} .

4. Throughput Collapse and the Role of Ψ

To operationalize sufficiency, we use **operational throughput**, denoted Ψ , following prior work on measurable throughput in open systems.

- Above F_{min} :
 Ψ degrades gracefully.
- At F_{min} :
The system operates at maximal efficiency.
- Below F_{min} :
 Ψ collapses abruptly.

This explains why pruning curves exhibit **phase-transition-like behavior**, analogous to:

- Percolation thresholds
- Rank collapse in linear systems
- Spectral support loss

Crucially:

Retraining cannot restore performance once F_{min} is violated, because the structural support required for inference no longer exists.

5. Pruning as a Sacrificial Necessity

Pruning is often framed as optimization or compression. We propose a different interpretation:

Pruning is a structural survival mechanism.

In effective systems, non-contributing substructures are actively shed to preserve coherence.

In machine learning, this manifests as:

- Magnitude pruning
- Block sparsity

- Neuron death during training

Efficient models survive not because they are sparse, but because they have already converged to a minimal sufficient structure.

6. Relation to the Lottery Ticket Hypothesis (Without Mythology)

The Lottery Ticket Hypothesis (LTH) demonstrated that small subnetworks can match the performance of full models when initialized correctly.

We reinterpret LTH as follows:

- LTH **describes discovery**
- F_{min} **explains necessity**

Winning tickets are not lucky subnetworks.

They are **instances of minimal sufficiency uncovered through overparameterization**.

Overparameterization enables exploration; pruning reveals what must remain.

7. Failure Modes

Common failure modes align naturally with violations of F_{min} :

- **Over-pruning** → loss of global modes
- **Structured pruning in the wrong basis** → breaks minimal structure
- **Late-stage pruning** → higher risk due to rigidified representations

These failures are not stochastic. They are structural.

8. Implications for Minimal-Resource Machine Learning

This framework yields direct implications:

- Training with **10–20% capacity** is feasible if structure is preserved.
- **Continuous learning** is more robust than static training followed by pruning.

- Hardware-aware design should target preservation of F_{min} , not maximal sparsity.
- Efficiency gains plateau once pruning approaches structural limits.

Minimal-resource ML is not about smaller models, but about **structurally sufficient ones**.

9. Conclusion

Neural networks do not fail because they lose parameters.

They fail when they lose structure.

Pruning succeeds when it removes expendable mass, and fails when it violates minimal sufficiency. Understanding this boundary is essential for the next generation of efficient machine learning systems.

Throughput vs Pruning Ratio Diagram

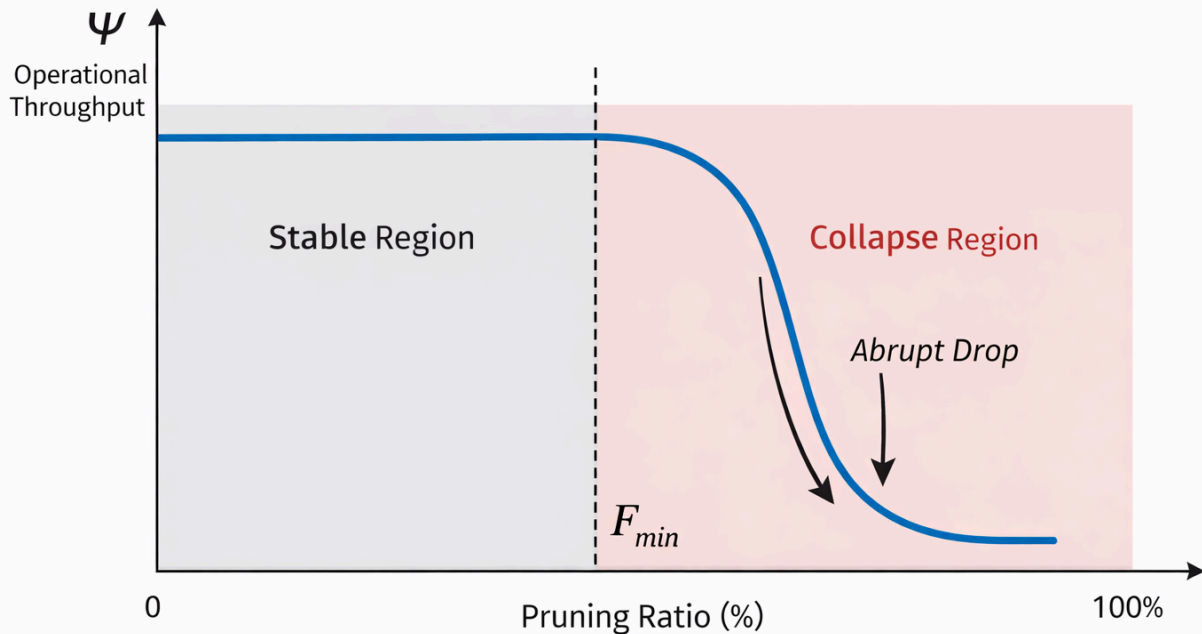


Figure 1: Throughput vs. Pruning Ratio

Figure 1 — Phase diagram of operational throughput under pruning.

The diagram shows the behavior of operational throughput (Ψ) as a function of the pruning ratio. For low to moderate pruning, the system remains in a *stable region*, where performance degrades smoothly or remains nearly constant despite substantial parameter removal. This regime corresponds to pruning that eliminates redundant or non-contributing substructures while preserving the system's minimal structurally sufficient core, denoted F_{min} .

The vertical dashed line marks the F_{min} boundary: the critical threshold below which the remaining structure is no longer sufficient to support stable inference. Once pruning crosses this boundary, the system enters the *collapse region*, characterized by an abrupt and non-linear drop in Ψ . In this regime, performance failure is structural rather than quantitative: additional training or fine-tuning cannot recover throughput because the minimal sufficient structure has been violated.

The diagram highlights that pruning success is not determined by sparsity level alone, but by whether the pruning process preserves F_{min} . Neural networks do not fail because they lose parameters; they fail when they lose the minimal structure required for coherent operation.

Appendix — Mapping to ISSP and Spectral Interpretation

This appendix connects the concept of **minimal structural sufficiency** ($F_{\min}$) with the **Induced Structural Stability Principle (ISSP)** and provides a spectral interpretation consistent with standard linear-algebraic views of neural networks.

A.1 ISSP: Structural Stability Under Constraint

The Induced Structural Stability Principle (ISSP) states that a system operating under external constraints converges toward a reduced set of structurally stable modes that dominate its functional behavior.

In machine learning systems, such constraints include:

- architectural boundaries (layer topology, block structure),
- hardware constraints (block sparsity, memory alignment),
- data-induced constraints (task structure, input manifold),
- training dynamics (regularization, optimization bias).

ISSP predicts that, under these constraints, functional behavior becomes concentrated in a limited subset of modes that are robust to perturbation. These modes define the **structural backbone** of the system.

Within this framework,

$$F_{\min}$$

corresponds to the **minimal subset of structurally stable modes required to sustain non-zero operational throughput** ($\Psi > 0$).

A.2 Spectral View of $F_{\min}$

Consider a trained neural network represented implicitly by an effective connectivity operator $\mathcal{W}$ (e.g., weight matrix, Jacobian, or linearized operator around typical inputs).

From a spectral perspective:

- The eigenmodes of $\mathcal{W}$ represent functional directions of information propagation.
- High-magnitude or well-conditioned eigenmodes dominate inference.
- Low-energy or weakly coupled modes contribute marginally or redundantly.

Under pruning or structural reduction:

- Non-contributing eigenmodes are removed with little effect on throughput.
- The dominant spectral subspace remains intact until a critical point.
- Once pruning removes or destabilizes one or more **dominant eigenmodes**, the system crosses the $F_{\min}$ boundary.

At this point, the spectral rank or effective condition number collapses, leading to an abrupt drop in Ψ .

Thus, $F_{\min}$ can be interpreted as:

the minimal spectral support required for stable inference under task constraints.

A.3 Relation Between ISSP, $F_{\min}$, and Ψ

The three quantities relate as follows:

- **ISSP** describes *why* systems concentrate functionality into a limited structural core.
- $F_{\min}$ identifies *what* that core is: the smallest structure that still supports coherent operation.
- Ψ measures *whether* the system remains operational.

Formally:

- Above $F_{\min}$:
Structural perturbations induce smooth changes in Ψ .
- At $F_{\min}$:
The system is marginally stable.

- Below $F_{\min}$:
ISSP predicts loss of dominant modes $\rightarrow$ rapid spectral degeneration $\rightarrow \Psi \rightarrow 0$.

This explains why performance collapse under pruning is abrupt rather than gradual.

A.4 Implications for Pruning and Efficiency

From this perspective:

- Pruning is successful when it respects the dominant spectral subspace.
- Sparsity alone is insufficient; **basis alignment matters**.
- Hardware-aware pruning succeeds when it removes redundancy while preserving structurally induced eigenmodes.

This framing unifies empirical observations from:

- Lottery Ticket Hypothesis,
- structured sparsity,
- early vs. late pruning asymmetries,
- irreversibility of some pruning failures.

All are consequences of violating—or preserving— $F_{\min}$ under ISSP.

A.5 Summary

ISSP provides the dynamical principle.

$F_{\min}$ provides the structural boundary.

Ψ provides the operational signal.

Together, they explain why neural networks tolerate massive reduction up to a point—and why crossing that point produces sudden, unrecoverable failure.

This closes the loop between pruning, structure, and throughput using standard tools familiar to machine learning and spectral analysis.